

July 14th – October 14th , 2025

CSM Training period –

HTML, CSS, Javascript, Typescript,
Postgres SQL, PGAdmin
Python, DataAnalytics, Data Visualization,
Machine Learning, Deep Learning,
Apache AirFlow, GenAI, Transformers, LLMs , Langchain,
GenAI
Module Assessments

October 2025

New Learnings

LangChain + (LangFlow and LangGraph)
Angular Frontend Framework and commands
Poetry Python package and commands

Projects – WhatsApp like frontend UI - Audio recorder for MedhaK
Using Angular Login pages
MedhaK LangGraph
JIMMS project

15-10-2025

Self Revision – DL, ML, Transformers, Loss, Optimizers, Metrics

16-10-2025

Team introduction,
Checking ChatBot dependencies,
Running the front and backend code
LangChain

17-10-2025

Self Learning on Angular
Angular and basic project
React + Graph amplifier

October 18th 2025

Saturday

October 19th 2025

Sunday

October 20th 2025

MONDAY (Diwali)

October 21st 2025

WhatsApp Like UI for Audio visualization
Frontend React – Visualization,
Discussion with Soumya regarding MedhaK chatbot Frontend

October 22nd 2025

Speech to Text,
React UI decoration(Like Whsaap Style),
Audio Recoding with stop and pause

October 23rd 2025

React UI decoration(Like Whsaap Style)
Editable Text
Medha.K Admin frontend – Dashboard, Users, Table and database , Keycloak,
Groups, permission, menu.

October 24th 2025

KT session on Attu and Milvus vector store
Self Practice on Attu vector store and (Project, Flows, DataRepository - Collections)

October 25th 2025

Saturday

October 26th 2025

Sunday

October 27th 2025

Angular Project – Login and Signup page

October 28th 2025

LangGraph and Simple Project
Peotry.lock and pyproject.toml
Peotry installation and setup
Simple random number generation code in “src\” and
testing code with pytest in “test\” for main code
KT discussion on Keycloak SSO

October 29th 2025

Disussion with Nitesh on logs, Project workspace
In langraph_crew_agent application,
Code workflow understanding (Done) :

- \ main.py,
- \ api \ routes.py,
- \ workflows \ langgraph_flow.py
- \ workflows \ state_model.py
- \ workflows \ nodes \ *.py
- \ agents \ crew \ sql_agent.py
- \ agents \ crew \ metadata \ schema_context.py
- \ agents \ crew \ tools \ sql_tool.py

\ agents \ crew \ core \ llm.py
Added try except error handling in main.py

October 30th 2025

Exception Handling Task - langgraph_crew_agent (Done and reported to Akash and Himalaya)

October 31st 2025

Transformers, Embeddings, Faiss DB

JIMMS project code workflow

Code understanding in : Acts and Rules.ipynb , Report_model_training.ipynb,
Report_model.ipynb

Discussion regarding next tasks from Monday and requirements with Dibyajyoti sir
Need to revise FastAPI, Sentence Transformers for Multi question for same answer.

November 2025

New Learnings

Model FineTuning and Sentence Transformers

Angular and FastAPI

PG Admin data crud operations – Frontend

ChatPage.js, SharedChat.js, SharedLinks.js

LangGraph, LangFlow, Frontend, Backend

November 1st 2025

Saturday

November 2nd 2025

Sunday

November 3rd 2025

Discussion – Fine Tuning and reports 5 categories ; Query as text or keyword.

Task (Done) – Backend api endpoints to load and get data

Frontend using api endpoints fetch and show the data

(Here, data = raw data json from JIMMS)

November 4th 2025

Jimms Project – Report and ActsRules .ipynb

Structure specific code adjustment in .ipynb for Report and ActsRules seperated

Added Evaluation metrics – steps vs loss

November 5th 2025

Research :

Fine Tuning and ST

Loss Function and Embeddings

Model FineTuning (Done) – Achieved Training loss 0.0006

FastAPI methods – GET, POST, PUT, PATCH, DELETE, OPTION

November 6th 2025

ChatBot speech – Audio TTS

PgAdmin table data on Frontend with crud operations

November 7th 2025

ChatBot speech – Audio TTS for English and filtered text ,

Ensuring audio by considering text and without symbols.

November 8th 2025

Saturday

November 9th 2025

Sunday

November 10th 2025

Adding design to shared_chat.js

Handling repeated saving to MongoDB while clicking on share button.

(flowbackendapi.py To be tested)

GIT commands for project

November 11th 2025

Changes and Updated - shared_chat.js and flowbackendapi.py

Documentation done for new changes

Change for handling duplicates data ,

Share chat and user information display

November 12th 2025

Laptop issue , fan changed

Understanding ChatPage.js

Updating in ChatPage.js for adding “generating...” feature

November 13th 2025

Added new SharedLinks.js file

Using SharedLinks.js file displaying a default message as of now

Add shared Links in settings to show all links associated with user

November 14th 2025

Modified SharedLinks.js, ChatPage files for displaying shared links on Frontend UI.

Created new <div> container for showing shared links instead of replacing a className="searchcontainer-div" <div> container.

Changes in FastAPI , flowbackendapi.py changes : unique_id is provided: return specific chat details ; unique_id is null/empty: return all shared links

November 15th 2025

Saturday

November 16th 2025

Sunday

November 17th 2025

Error Fix while chatbot login
Back Button alignment right corner
Add delete button for each shared link url

November 18th 2025

Frontend Changes for displaying shared information

November 19th 2025

Frontend and Backend changes for UI and delete/delete all Links

November 20th 2025

Data format , Auto reload for Share Link
Frontend buttons position and styling
Unique_id and Unique_que_id

November 21st 2025

Version 2
Admin Backend, Frontend
Bot_workflow Backend, Frontend
Chatbot

November 22nd 2025

Saturday

November 23rd 2025

Sunday

24-11-2025

Bot workflow New button functionality for creating new flow

25-11-2025

Worked on Admin Frontend for Tag button , popup is coming but flow names not loaded.

26-11-2025

Worked on tag button , able to Tag the flow,
Renaming the button names.

27-11-2025

Langgraph code R&D and understanding
Created the API Node

28-11-2025

Hide Settings side panel
Confirm window popup for delete flow
Updated the Date and Time but flow name again not updated

December 2025

New Learnings

RAG with LangChain and LangGraph NodeRegistration in Frontend

01-12-2025

Delete button confirmation popup from new NotificationPopup.tsx file (Done)

Created at and modified at for each flow (Done)

Header and search bar freeze (Done)

Added Import and Export button in Header.tsx (Functionality need to implemented)

02-12-2025

Added Import and Export button in Header.tsx

Debugging git push updated code

Code cleanup

Flows size fixing while search

03-12-2025

'Delete' as option for each node instead of 'x'

Hide options by clicking anywhere

Background white remove for output node

Padding border remove for drag drop nodes

04-12-2025

Frontend change for drop option,

Styling changes for drop option,

Node Registry changes

05-12-2025

API Node drag and drop feature,

Code cleanup,

Updated type.ts and nodeRegistry.ts

06-12-2025

Saturday

07-12-2025

Sunday

08-12-2025

3 dots in drag drop Node.

Close all drop down menu option by default.

R&D

Search component for nodes <div> declaration functionality pending
pipeline and pipelinev2 in Dropdown menu for AI Components

09-12-2025

Added new nodes for MCP - server and client.

UI design and code clean-up

10-12-2025

On Leave

11-12-2025

Creating new sub-node for LLM

Created Prompt sub node

12-12-2025

Setting up condition for sub-node for restricting the connections

Dropdown option preservation

Hide scroll bar for prompt sub node

13-12-2025

Saturday

14-12-2025

Sunday

15-12-2025

API KEY abstract

added temperature

Backend with Sangram for LLM node

16-12-2025

LLM node test (API key + model) successful

Preseving dropdown option even after refresh

In Fronted FaissDB – Threshold and DB path added.

Added FaissDB Node and need to implement in Backend

17-12-2025

FaissDB node preserve values

Human in loop

Created nodes for Email, Event Listen , Approval Node,

Testing Approval Node,

18-12-2025

FaissDB (Done)

Approve Node (Done)

SubGraph node added , getting flow name but cant preserve (In Progress)

19-12-2025

Preserve the saved flows (Done)

Flow edit run (Reviewed, Done)

20-12-2025

Saturday

21-12-2025

Sunday

22-12-2025

Merging and resolving conflicts – Akash ,Bot workflow backend and frontend (Done)

Changes in files for UI remove cross “X” button after tagging Flow (Done)

23-12-2025

Vector DB – Faiss and Retrieval (Done)

Agent – BI Agent (Done)

NODE_REGISTRY and NODE_CATEGORIES break connection (Done)

For “latest progress” Run taken backend code from “Sangram_llm” branch.

24-12-2025

Node LangFlow Edge delete, Specific edge connection (Done)

NodeRegistry.ts to split folder (Done)

25-12-2025

HOLIDAY

26-12-2025

Dropdown for Pipeline FlowID (Done)

Backend pull Sangramkp (Done)

MongoDB service installed (Done)

HIL Approval (InProgress) –

Approval pop came

Output+Approve+Deny Buttons Popup came

Due to backend cant resume the Flow after Approve button

Dheeraj_Process_Test tested with Dummy used APIkey and got dummy results

Dheeraj_llm_Test tested – Got response

Helpdesk tested – Got response

Git push done in both Backend and Frontend.
Will send merge request once it get reviewed.

27-12-2025

Saturday

28-12-2025

Sunday

29-12-2025

DB_shcema_2(Done, To Be Reviewed)

Understanding DB schema for flows(Done)

Git push and merge request for pipelinev2 dropdown for flow_v2(Done, TBR)

30-12-2025

All Code and functionality tested and reviewed (Done)

Load flow effect after Delete flow (Done)

Approval Testing (Done)

Git Push and Git merge in Queue (Done)

31-12-2025

Documentation for Node registry and node creations (In Progress)

0.5 ML second half

January 2026

New Learnings

API intergration testings
Flow execution testings and debugging
RAG from scratch
Transformers architechture for AI platforms
Backend structure

01-01-2026

HOLIDAY

02-01-2026

API testing with Sangram : API + Approval test
Documentation for Node registry and node creations + Images load (Done)

03-01-2026

Merge nodes (To be reviwed and Tested done)
Frontend similar to <http://192.168.10.208:7861/> Design (To be reviewed)
Dark and light theme (In Progress)

04-01-2026

Sunday

05-01-2026

Frontend similar to <http://192.168.10.208:7861/> Design (To be Final Reviewed)
+ button for adding a node on UI (Done)
Dark and light theme (Done)

06-01-2026

CSMS Frontend plan (Reviewed need changes)
CDMS backend code implementation and testing (In progress)

07-01-2026

Home and Back button navigation routing in Admin to Botworkflow, vice versa
(Reviewed need changes)
DB saving for uploaded files in CDMS test and disabled this feature (Done)

08-01-2026

Back Home Button navigation (Done)

Language detection Odia, English , Roman Odia (Done)
Docker training (In progress)

09-01-2026

Git pull main , merge conflicts (Done)
Chart Bot Workflow with Admin Backend (In Progress)

10-01-2026

Saturday

11-01-2026

Sunday

12-01-2026

Auto fit view after DragDrop or add node or refresh (Done)
Response from Admin Backend , Chat response in Bot (Done)

13-01-2026

confiurl + frontend + ws url dynamic
Workflowid dynamic in chatb page from admin backend
Docker training

14-01-2026

On Leave

15-01-2026

On Leave

16-01-2026

On Leave

17-01-2026

Saturday

18-01-2026

Sunday

19-01-2026

WorkflowID in frontend and tested a flow having workflowID.
working on AI flow Execution Pipeline V2 + LangflowAPI key

20-01-2026

Git pull main (Done)
In .env file removed /advanced from flow_v2_response_url (Done)
Getting responses Admin + Chatbot. (Done)
Added new node : "Suggestion Node"(Done)
Added new functionality for getting checked items in output node. (Done)

21-01-2026

ChatPage.js modification for getting and displaying all nodes status while chat Tested the bots – i3msTestagent and HR assistant (Done)

22-01-2026

New node for python fuction + Dropdown function names (Done)
ChatServices.py and similaritySearch function with flowID (Done)

23-01-2026

RAG with Langgraph standalone code by R&D (Done)
New Pipeline Flow created and tested with Python function , Suggestion and AI flow piplinev2 Nodes execution. (Done)

24-01-2026

Saturday

25-01-2026

Sunday

26-01-2026

Holiday

27-01-2026

Working code for pipeline : Python function, Suggestion, Merge, AI flow Execution (Pipeline v2)
Discussion about Agent and upcoming features

28-01-2026

New node creation Agent Node, Frontend and backend code implementation for fetching data from database to show in dropdown.

29-01-2026

Agent Node for API endpoint and dropdown for agent name and execution name

30-01-2026

AgentID bind for agent node (Done)
I3ms flow test (Done)
Suggestion node UI changes name,text,opt_id aka “pk”(Done)
Run button disable while execution (Done)

31-01-2026

Saturday

FEBRUARY 2026

New Learnings

Backend workflow and project structure
Backend and executionservices
botworkflow project structure and code
LLM and SubNodes and Graphbuilder
Code R&D execute endpoint
LangChainOpenAI structured output

01-02-2026

Sunday

02-02-2026

Enable Run button - Close 'X' button in Suggestion panel UI. (Done)
Git pulled main and executed the i3MS flow. (Done)
R&D on AI Transformers
Api key from .env for AI Flow Execution pipelinev2 (Done)

03-02-2026

New node creation - Parser Node (Done)
Parser Node functionality added and tested , similar to Python node. (Done)
Output manipulation for getting output in Specific required format (Inprogress)

04-02-2026

R&D on full backend and frontend codes of botworkflow – New features for input manipulation
Parser node testing

05-02-2026

Fixing parser node data transfer to frontend

06-02-2026

requirements.txt exact version fix
Home button replacing with csm logo
Suggestion node in I3ms

07-02-2026

Saturday

08-02-2026

Sunday

09-02-2026

I3MS flow test

Frontend changes

10-02-2026

1)browser title= CSM Ai Foundary

2)Remove -Bi Agent

3)Agent rename to = Inbuilt Agent

4)tools ==> python function

5)remove langdetection

6)csv loader description check

7)loader rename to ==> Data Injection

8)Prompt rename to prompt template and move to ai components

9)Ai components==> sub graph

10) adding guardrail menu (less priority)

I3MS flow test

11-02-2026

nodeid to nodelabel in execution panel.tsx

user input in mongodb

frontend change for light theme logo

12-02-2026

Chatbot + latest working i3ms_v2 + chatbot

13-02-2026

Redis Node + latest working i3ms_v2 + chatbot

14-02-2026

Saturday

15-02-2026

Sunday

16-02-2026

On Medical Leave

17-02-2026

Edge and sub node frontend changes and tested

Edge + label
Parser node function fix backend

18-02-2026

Parser node fix
Mail Node functionality
R&D on Pydantic, TypedDict, Dict, JSON

19-02-2026

Git pull main and working latest branches for all 5 v2 projects – botworkflow, admin, chatbot
Self R&D and self implementation without GPT reference : FastAPI + React sample project.

20-02-2026

New node creation and adding functionality – Structured output node
Tested for 3 types – Pydantic, JSON and TypedDict

21-02-2026

Saturday

22-02-2026

Sunday

23-02-2026

LLM + 2 subnodes (LLM + Structure output LLM)

24-02-2026

LLM + 2 subnodes (LLM + Structure output LLM)
Chatbot sending node labels

25-02-2026

LLM + 2 subnodes (LLM + Structure output LLM)
And Tested the LLM + Structure output LLM

26-02-2026

ChatPage NodeLabel during processing and executing...
Git Pull state_variable Frontend New Changes

27-02-2026

R&D on state_variable Frontend

28-02-2026

Saturday

March 2026

New Learning

StateVariable and Node mapper functions
Input and Output mapping
Dynamic LLM, I3MS and Agent flow testing + Tools

01-03-2026

Sunday

02-03-2026

R&D on state_variable Backend

03-03-2026

R&D on LangGraph
Exploring AI agents and Agentic AI

04-03-2026

HOLIDAY

05-03-2026

Skeleton effect for each node
Current and Previous node execution

06-03-2026

R&D on new updated code and LangGraph
LLM Node and Prompt Node + State Graph

07-03-2026

Saturday

08-03-2026

Sunday

09-03-2026

R&D on usage of XML file and BaseNodeMapper class.
XML Input and Output mapping with Agent Node .

10-03-2026

R&D on usage of JSON file and BaseNodeMapper class and NodeMapper().

LLM model + Subnodes

11-03-2026

Discussion with team regarding Botworkflow overview and abouts

Tooltip at edge

Skeleton effect while node active

12-03-2026

Skeleton and tooltip (Done)

R&D on Frontend debugging

R&D on Backend for StateVariable

13-03-2026

WorkflowConfigModal.tsx loading effect

<StateVar>.<Key> fixed in base_node.py in backend

ReducerType Validation and cheking

14-03-2026

Saturday

15-03-2026

Sunday

16-03-2026

Suggestion Node development

Approval Node development

17-03-2026

Approval Node development

Approval feedback

Import button fixed duplicate flow was creating

18-03-2026

Suggestions from Python Function I3MS_parser funtion

Workflow Config state variables UI

Flow I3MS test with StateVariable

Parser output function pending

19-03-2026

On Leave

20-03-2026

AI flow execution to LLM Flow

Redis Node visibility displaying
Data Injection to Data Connector renaming
Inbuilt Agent in AI Components shifted

Retrieval (New Node Frontend)
- Milvus Retriever

21-03-2026

Saturday

22-03-2026

Sunday

23-03-2026

Git pull state_variable_akash
LLM Node Test after resolving git pull conflicts
Agent Node demo discussion
Agent Node test after resolving git pull conflicts

24-03-2026

LLM Node fixing
state variable node type display

25-03-2026

Node variable node type display
Dynamic mapping fixing for LLM Nodes

26-03-2026

Redis in Action Shifted
Agent to Models Renamed
Chroma, Milvus and Faiss in Vector Database
Dynamic mapping fix after the git pull

27-03-2026

Assisting, guiding and implementing : Tools \ *.ts with Dibya in creating new nodes
and testing

28-03-2026

Saturday

29-03-2026

Sunday

30-03-2026

KT on new version – Last week + 2 ver push

Workflowconfig.tsx drag and flex

Need New version that can fix Dynamic mapping and Flow run for new nodes

Found edge starting at unknown node 'ReactAgent_1'

31-03-2026

R&D on GraphBuilder and BasenodeMapper

Changes GraphBuilder and BasenodeMapper

Tesing flow

April 2026

New Learning

AI agent tools
ai.dev and vide coding tool
Langchain and Langgraph Tools
Clearing python cache - “pip cache purge” command
Sarvamai for TTS ans STT
.tk and sso for free domain

01-04-2026

Streamingpopup.tsx to show the chunks in typing effect Final result + Execution logs
R&D on Frontend code - Botworkflow

02-04-2026

R&D on Frontend code - Botworkflow
Streamingpopup.tsx to show the chunks in typing effect for each node + Execution logs

03-04-2026

Updated the get_updates() in base node for llm node outputmapping. TBD,TBT

04-04-2026

Saturday

05-04-2026

Sunday

06-04-2026

R&D on new git pull main code
Main code MongoDB to Redis and redis to Mongo for optimal and fast execution

07-04-2026

changes in /execute endpoints
Redis operation in State Variable

08-04-2026

Redis operation and configuration set as per 7861 and 7865 ports
Flow testing ensuring redis working properly or not.

09-04-2026

I3MS flow and AMP BI flow fix
Redis code R&D

StateVariable code summay log show

10-04-2026

main.py changes for logging in Mongo
Redis code brach stable

11-04-2026

Saturday

12-04-2026

Sunday

13-04-2026

SV_Dheeraj_1 for start node output return
Via function TBD

14-04-2026

Support and testings for new bot and flow SWS chatbot with Pallabi mam and Ollama models.

Fixed duplicate binding in execution summary.

llm_node dynamic binding and need to bind on UI field

15-04-2026

Exec_Dheeraj_1 : Question Bind and insert chat_history

16-04-2026

SV_Dheeraj_1 : Question Bind + Final output on UI

SV_Dheeraj_1 + akash_dev

17-04-2026

SWP flow test + Redis + Milvus

Git pull akash_dev

Frontend and backend changes

18-04-2026

Saturday

19-04-2026

Sunday

20-04-2026

If else, milvus ,retrival

testing chatbot

testing for MongoDB

21-04-2026

Merging SV and Main and testing chatbot

22-04-2026

Stable version testing for merged code (Main + State Variable)

Fixing and support to the team for SWP connection

23-04-2026

Fixing and support to the team for SWP connection

Node label binding in chatbot application.

24-04-2026

SessionThreadid fix and run to get threads

Thread Switch

Stable version update and SWP flow testing

25-04-2026

Saturday

26-04-2026

Sunday

27-04-2026

Export and Import button for WorkflowConfig

Effect when node is selected when side bar is open for selected node

Milvus Node (Remove : uri, user,pwd and implement using credential Node)

28-04-2026

Playground and Clear Cache buttons visible will get hide when Node is selected.

When Node selected - Hover effect for selected node

- Hide Playground and Clear Cache buttons

Milvus Node and Retrieval Node fixed and Tested the flow

29-04-2026

Copy and download button for results in Executions panel UI.

Copy and download button for chat play in Playground UI.

Select the text Area or Text Box then large UI for text area should open for seeing full value for it. And need a confirm button

Cursor enable in Frontend

30-04-2026

New Node functionality and testing – Authenticator node

MAY 2026

01-05-2026

May Day – Holiday

02-05-2026

Saturday

03-05-2026

Sunday

04-05-2026

Authenticator Node Test with ChatBot

05-05-2026

CIMS flow implementation

06-05-2026

CIMS flow Test

Fixing Retrieval node to fetch docs

07-05-2026

CIMS flow Test

Fixing the <StateVar>.<Key> : Get by key for JSON

08-05-2026

ON LEAVE - SL

09-05-2026

Saturday

10-05-2026

Sunday

11-05-2026

Stable version till date (Git version and Flow testing)

R&D on AI Node and New nodes

12-05-2026

R&D on Agent and Langgraph Tools

LangChain and Langgraph

13-05-2026

GenAI topics and discussion

Langchain and LangGraph scratch project(backend + frontend)

14-05-2026

R&D on GenAI topics and discussion

Langchain and LangGraph (RAG)

R&D on NPR and OCR model engine

Demo KT on Node Creation

R&D on Sarvamai node

New Node - Converter node